

Lab experiment-5

Experiment 5: Missionaries and Cannibals Problem

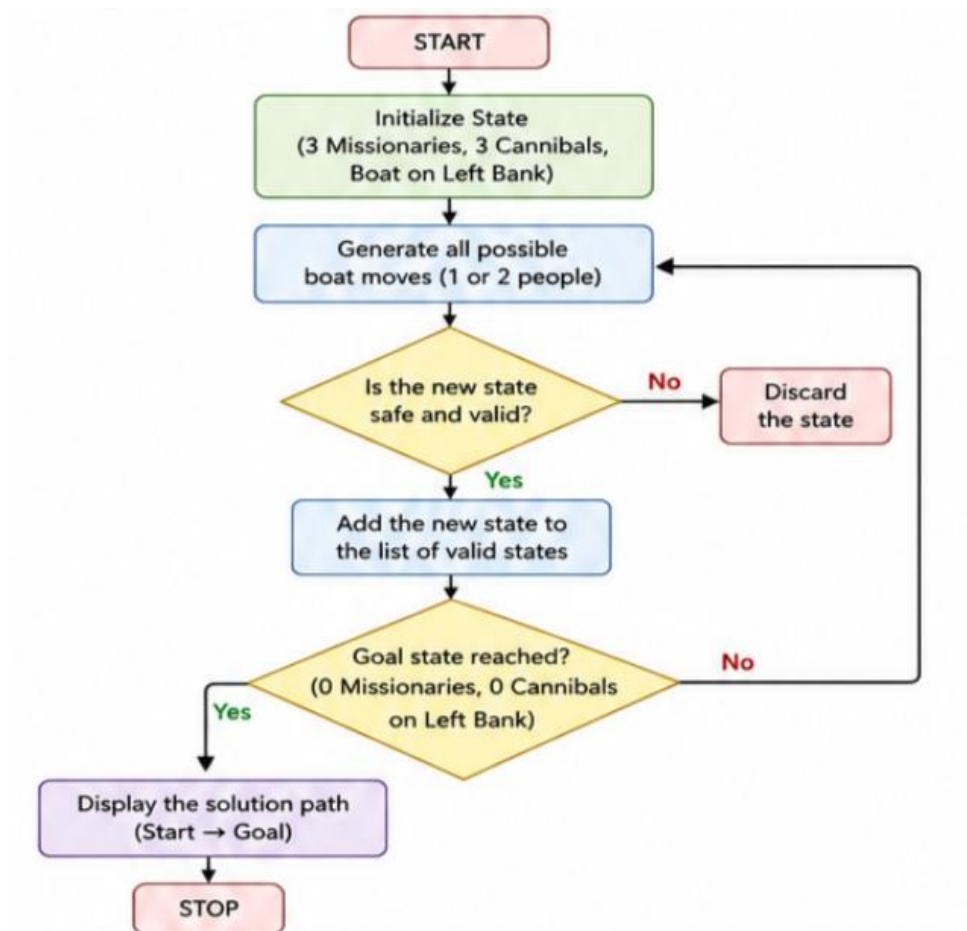
Aim

To develop a Python program to solve the Missionaries and Cannibals problem using State Space Search.

Objective

To transport three missionaries and three cannibals safely across the river without allowing cannibals to outnumber missionaries on either bank.

Flowchart



Algorithm

1. Initialize the starting state (3 missionaries, 3 cannibals, boat on left).
2. Generate all possible boat moves.
3. Check whether each new state is safe.
4. Ignore invalid states.
5. Continue exploring valid states until the goal state is reached.
6. Display the solution path.

Code

```
from collections import deque
```

```
def is_safe(m, c):
```

```
    if m > 0 and m < c:
```

```
        return False
```

```
    if (3 - m) > 0 and (3 - m) < (3 - c):
```

```
        return False
```

```
    return True
```

```
def missionaries_cannibals():
```

```
    start = (3, 3, 1)
```

```
    goal = (0, 0, 0)
```

```
    queue = deque([(start, [])])
```

```
    visited = set()
```

```
    while queue:
```

```
        (m, c, boat), path = queue.popleft()
```

```
        if (m, c, boat) == goal:
```

```
            return path + [(m, c, boat)]
```

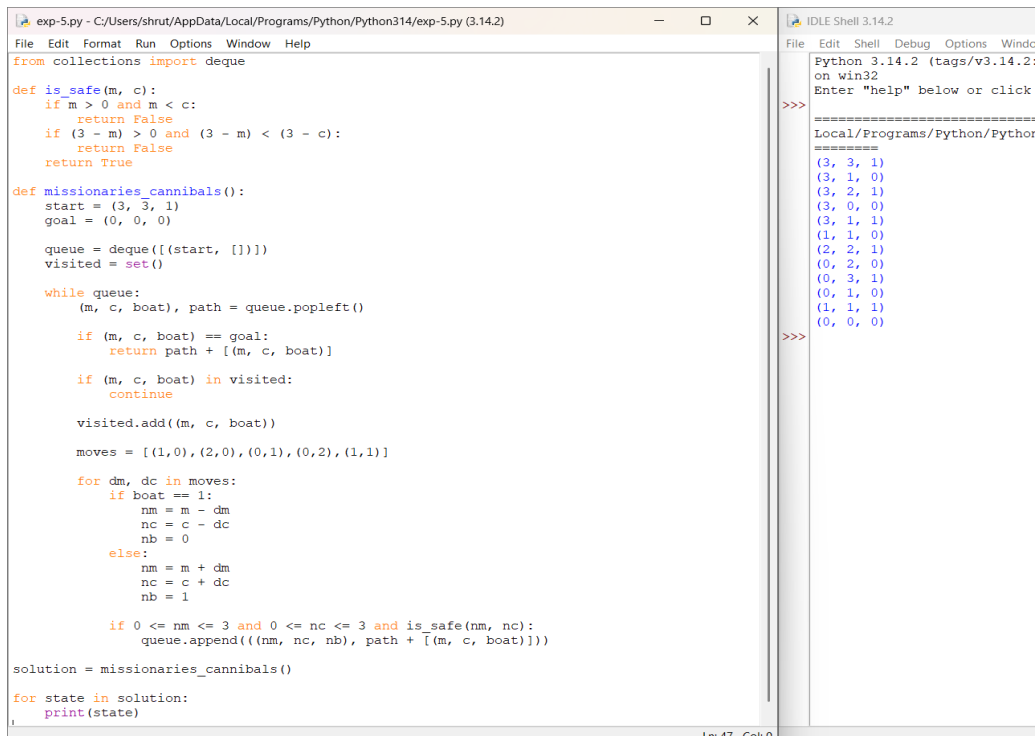
```
if (m, c, boat) in visited:
    continue
visited.add((m, c, boat))
moves = [(1,0),(2,0),(0,1),(0,2),(1,1)]
for dm, dc in moves:
    if boat == 1:
        nm = m - dm
        nc = c - dc
        nb = 0
    else:
        nm = m + dm
        nc = c + dc
        nb = 1
    if 0 <= nm <= 3 and 0 <= nc <= 3 and is_safe(nm, nc):
        queue.append(((nm, nc, nb), path + [(m, c, boat)]))
```

```
solution = missionaries_cannibals()
```

```
for state in solution:
```

```
    print(state)
```

Output



The screenshot shows a Python IDE with two windows. The main window displays a Python script for solving the Missionaries and Cannibals problem. The script defines a function `is_safe` to check if a state is valid, a function `missionaries_cannibals` to find the solution path, and a loop to print the solution. The output window shows the execution of the script, displaying the initial state, the goal state, and the sequence of moves that lead to the solution.

```
exp-5.py - C:/Users/shrut/AppData/Local/Programs/Python/Python314/exp-5.py (3.14.2)
File Edit Format Run Options Window Help
from collections import deque

def is_safe(m, c):
    if m > 0 and m < c:
        return False
    if (3 - m) > 0 and (3 - m) < (3 - c):
        return False
    return True

def missionaries_cannibals():
    start = (3, 3, 1)
    goal = (0, 0, 0)

    queue = deque([(start, [])])
    visited = set()

    while queue:
        (m, c, boat), path = queue.popleft()

        if (m, c, boat) == goal:
            return path + [(m, c, boat)]

        if (m, c, boat) in visited:
            continue

        visited.add((m, c, boat))

        moves = [(1, 0), (2, 0), (0, 1), (0, 2), (1, 1)]

        for dm, dc in moves:
            if boat == 1:
                nm = m - dm
                nc = c - dc
                nb = 0
            else:
                nm = m + dm
                nc = c + dc
                nb = 1

            if 0 <= nm <= 3 and 0 <= nc <= 3 and is_safe(nm, nc):
                queue.append(((nm, nc, nb), path + [(m, c, boat)]))

    solution = missionaries_cannibals()

    for state in solution:
        print(state)
```

```
Python 3.14.2 (tags/v3.14.2:
on win32
Enter "help" below or click
=====
Local/Programs/Python/Pythor
=====
(3, 3, 1)
(3, 1, 0)
(3, 2, 1)
(3, 0, 0)
(3, 1, 1)
(1, 1, 0)
(2, 2, 1)
(0, 2, 0)
(0, 3, 1)
(0, 1, 0)
(1, 1, 1)
(0, 0, 0)
>>>
```

Result

The Missionaries and Cannibals problem was successfully solved.

Conclusion

The program safely transported all missionaries and cannibals without violating the constraints.